SQL BASICS

Data Exploration and Visualisation

2025/2026/2

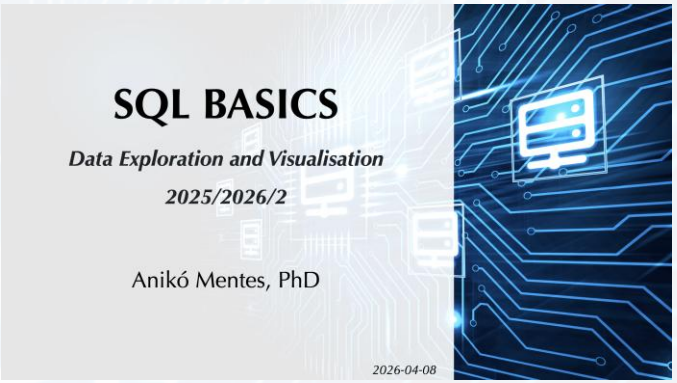
Anikó Mentes, PhD

2026-04-08

Today’s plan

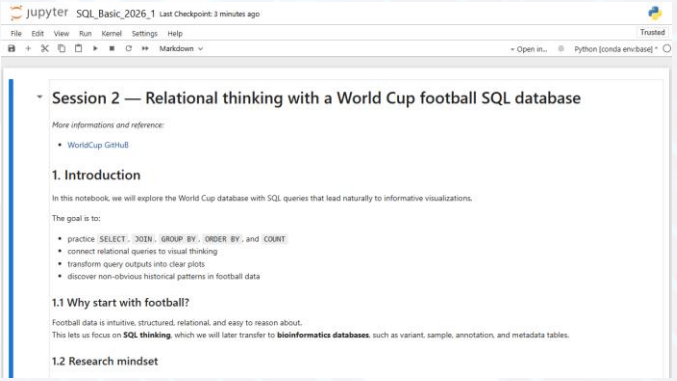
1. **Basics of Databases and SQL**

   basic_sql.pdf



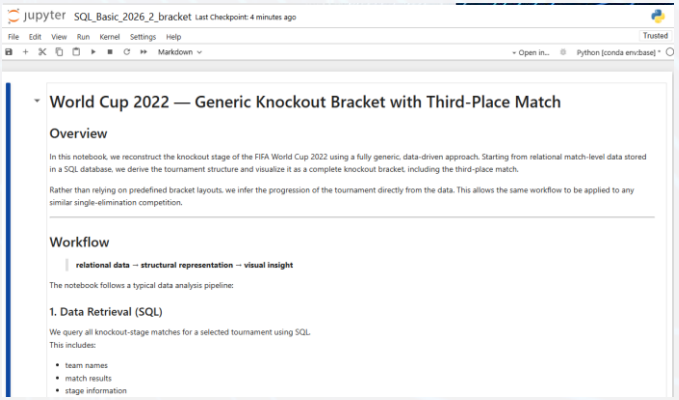
2. **Soccer WC database – examples**

   SQL_Basic_2026_1.ipynb



3. **Build your own Tournament Bracket – pipeline**

   SQL_Basic_2026_2_bracket.ipynb



History of databases

- Foundations of data thinking:
  - ***Aristotle*** (4th c. BCE): classification of entities
  - ***Plato*** (4th c. BCE): abstraction of reality
  - ***Charles Sanders Peirce & Gottlob Frege*** (19th c.): formal logic

- **Modern databases = formal representation of relationships**

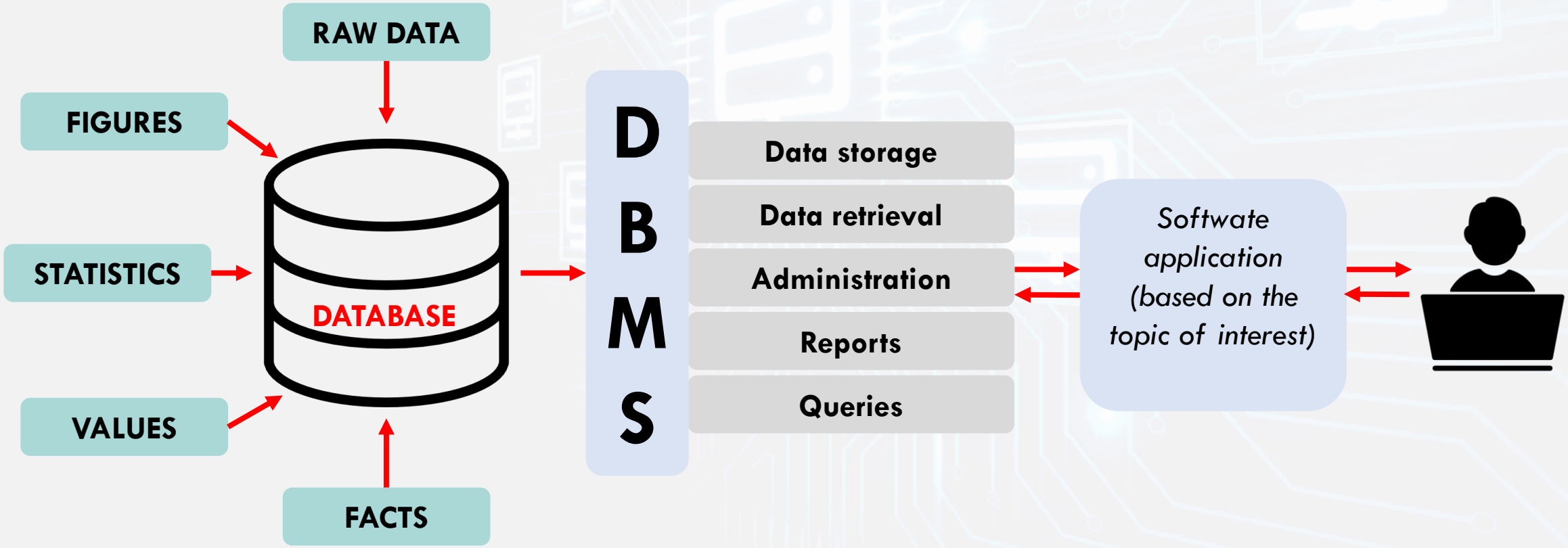
1960s	1970-80s	1990s-2000s	2005+	2015+
File Systems	Relational DB	Data Warehouses	NoSQL	Cloud & AI
Flat files	Tables + schema	OLAP	Flexible schema	Serverless
No queries	SQL	Indexes	Distributed	Real-time
Tight coupling	ACID	Optimization	Scalable	AI queries

What is DBMS?

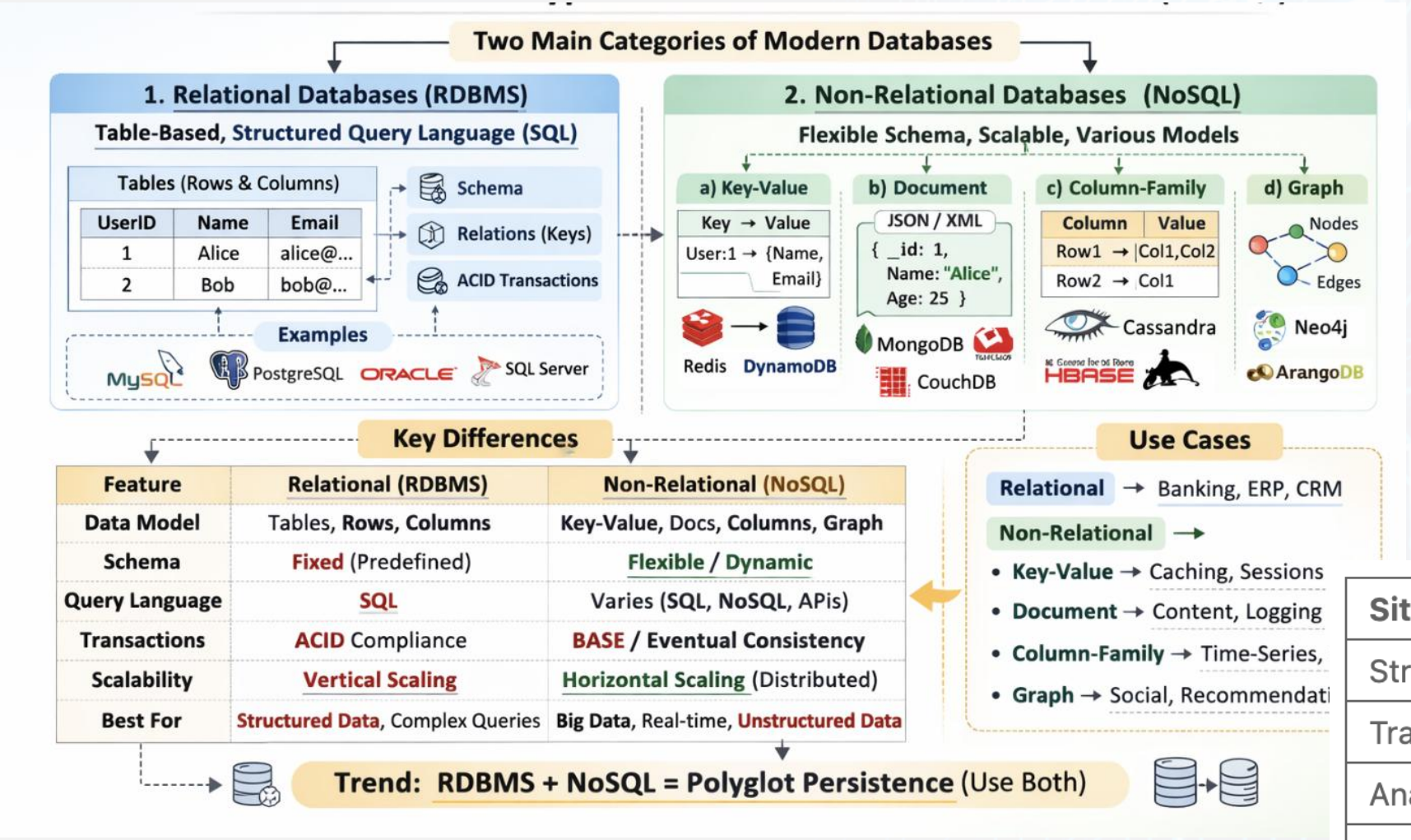
Data → Database → DBMS

- **Data:** raw facts (numbers, text, measurements)
- **Database:** organized collection of related data

Data**B**ase **M**anagement **S**ystem: Software package that enables: storage, quering, updates consistency and manage data in DB



Modern database types



Situation	Good choice
Strong relationships	Relational
Transactions	Relational
Analytics	Relational
Massive unstructured logs	NoSQL
Flexible documents	NoSQL

Which DBMS is used by the most in 2026?

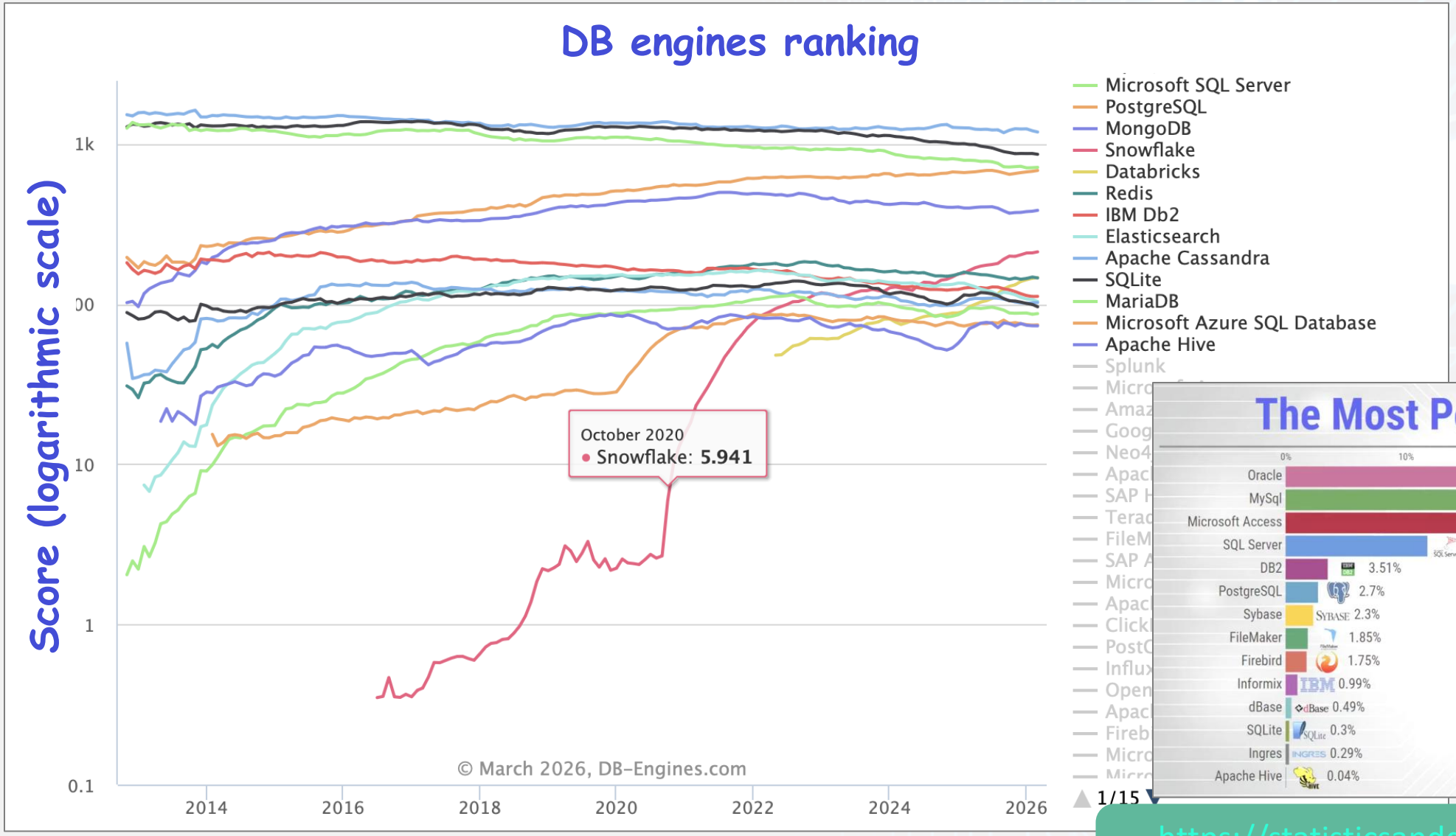
429 systems in ranking, March 2026

Rank			DBMS	Database Model	Score		
Mar 2026	Feb 2026	Mar 2025			Mar 2026	Feb 2026	Mar 2025
1.	1.	1.	Oracle	Relational, Multi-model	1182.46	-21.05	-70.62
2.	2.	2.	MySQL	Relational, Multi-model	858.34	-9.88	-129.79
3.	3.	3.	Microsoft SQL Server	Relational, Multi-model	711.47	+3.33	-76.67
4.	4.	4.	PostgreSQL	Relational, Multi-model	680.08	+8.05	+16.66
5.	5.	5.	MongoDB	Document, Multi-model	383.58	+4.85	-12.85
6.	6.	6.	Snowflake	Relational	211.24	+3.10	+49.46
7.	↑ 8.	↑ 13.	Databricks	Multi-model	145.81	+1.29	+49.80
8.	↓ 7.	↓ 7.	Redis	Key-value, Multi-model	145.19	-1.85	-10.17
9.	9.	9.	IBM Db2	Relational, Multi-model	111.38	+0.16	-15.19
10.	10.	↓ 8.	Elasticsearch	Multi-model	103.58	-2.88	-27.80

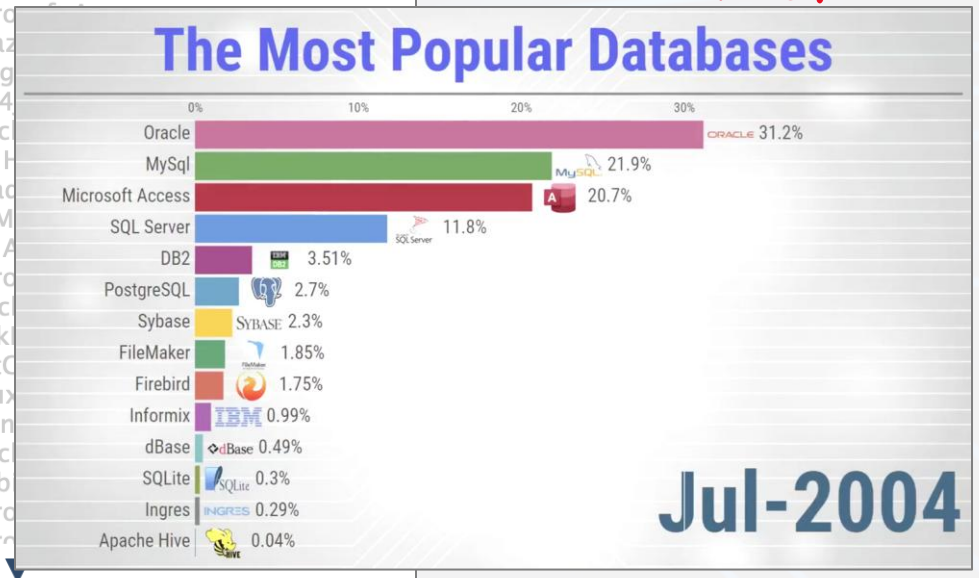
RDBS

https://db-engines.com/en/ranking

Which DBMS is used by the most in 2026?



Watch the video



https://db-engines.com/en/ranking_trend

https://statisticsanddata.org/data/the-most-popular-databases-2006-2023/

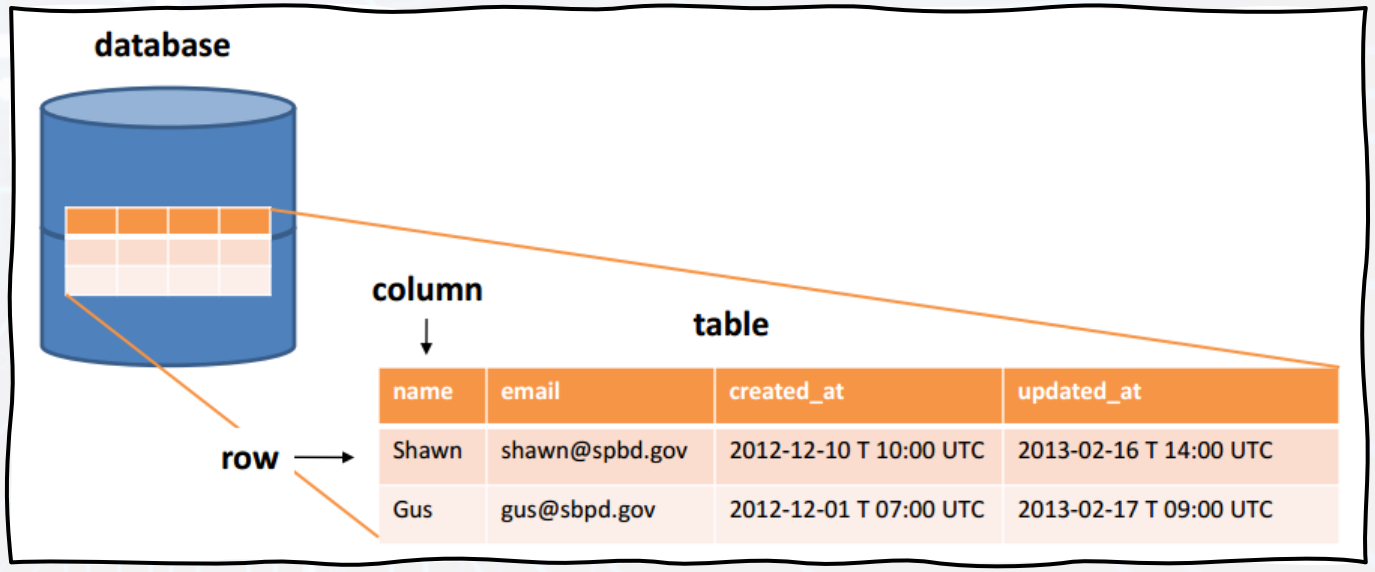
The relational model

Relational databases were introduced in the 1970s.

Key ideas:
- Data is stored in **tables** (relations)
- Tables consist of **rows** (tuples) and **columns** (attributes)
- Each table has a defined **schema**

Example structure:
- Table name
- Column names and types
- Constraints

A schema defines:
- the structure of the database
- the types of data stored
- relationships between tables



! The relational model provides a formal and structured way to represent data.

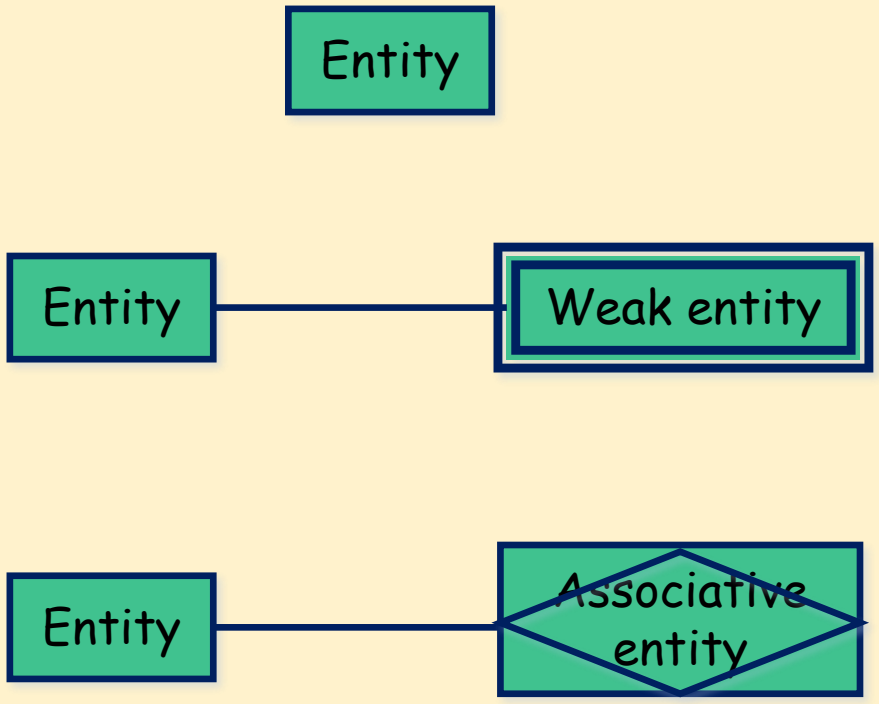
ACID

All RDBMS are ACID-compliant: Atomicity, Consistency, Isolation, and Durability.
- Data modification consisting of multiple steps
- Many transactions can be done concurrently

- **Atomicity**
  Transaction either finishes completely or nothing happens at all
- **Consistency**
  Transactions can take the database from consistent state to another consistent state only
- **Isolation**
  Concurrently running transactions might interfere with each other but only up to a certain, user-defined limit
- **Durability**
  Once a transaction is reported to be complete, it cannot be rolled back

Entity Relationship Model (ERD)

- **ERD:** a visual representation of different data using conventions that describe how these data are related to each other

Components and features	Example representation
Entity: real-world object, attributes give them their identity *Entity-set: collection of similar types of entities* *Weak entity: depends on the existence of another entity*	
Attributes: entities are represented by means of their properties *Types: Simple, composite, derived, single-value, multi-value, stb*	
Relationships: diamond-shaped box, Degree of relationship: Binary = degree 2, ternary = degree 3, n-ary	

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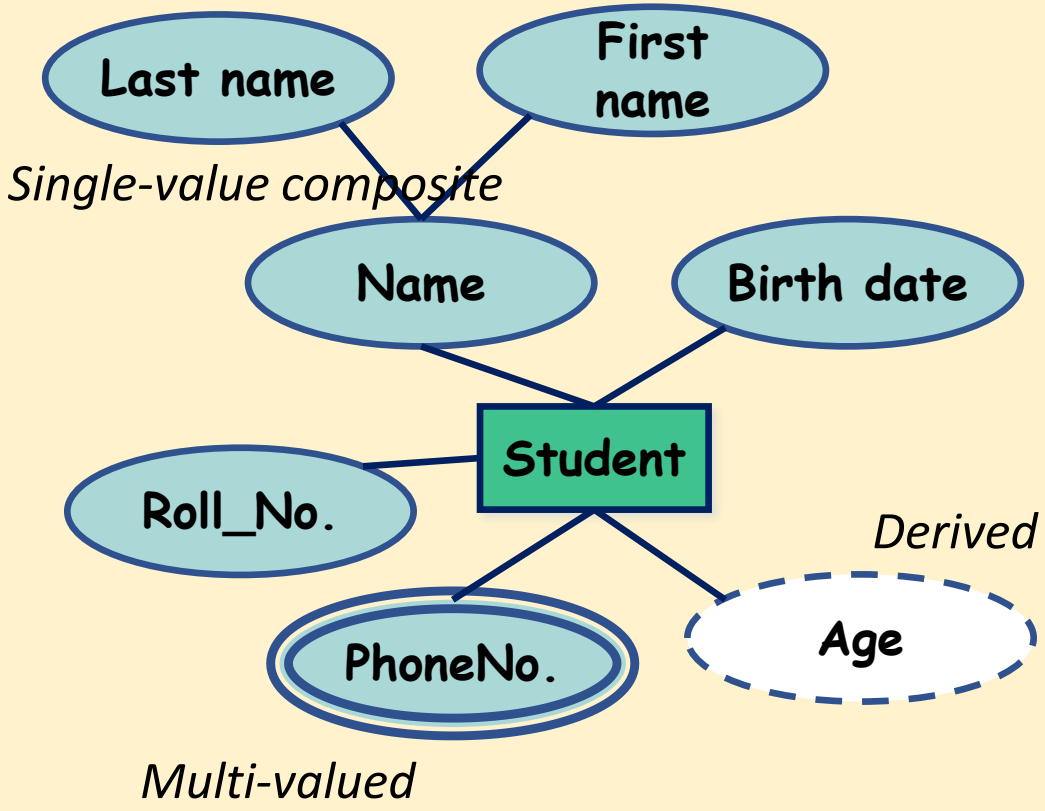
Entity: real-world object, attributes give them their identity Entity-set: collection of similar types of entities Weak entity: depends on the existence of another entity	
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Entity Relationship Model (ERD)

Entity keys: *Refers to an attribute that uniquely defines an entity in an entity set.*

Type	Description
Super	*A set of attributes (one or more) that together define an entity in an entity set.*
Candidate	*A minimal super key, meaning it has the least possible number of attributes to still be a super key. An entity set may have more than one candidate key.*
Primary	*A candidate key chosen by the database designer to uniquely identify the entity set.*
Foreign	*Identifies the relationship between entities.*

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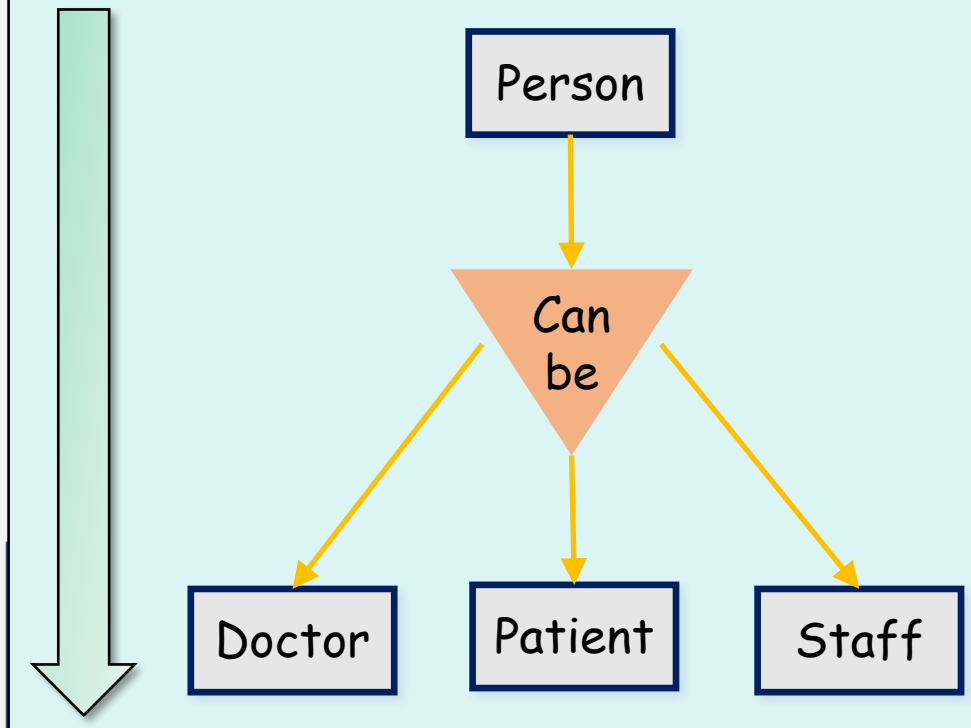
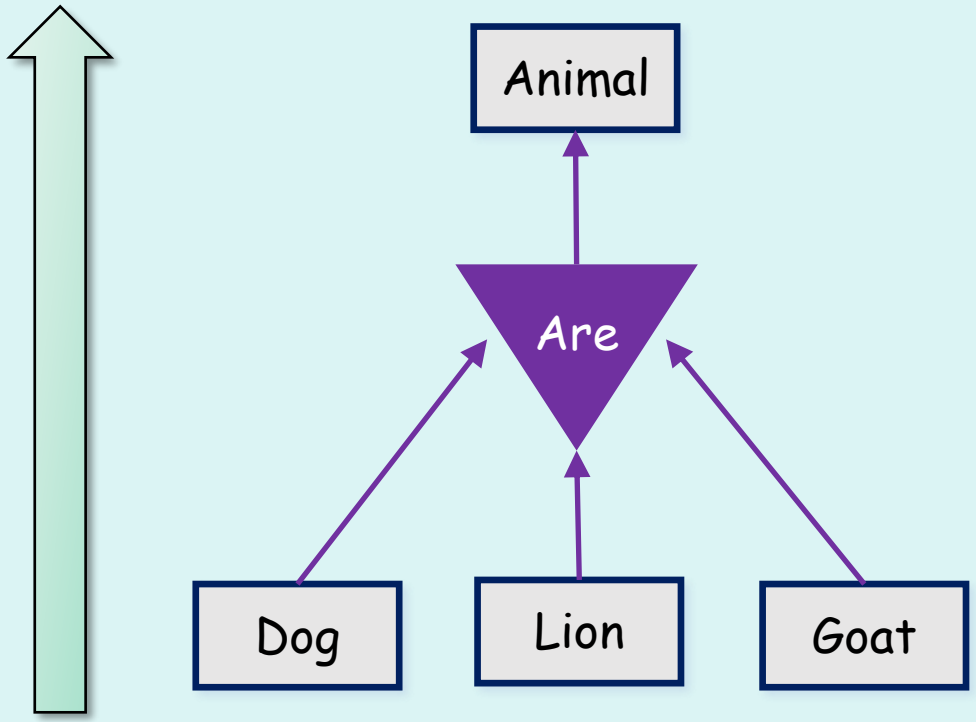
Components and features	Example
Entity: real-world object, attributes give them their identity *Entity-set: collection of similar types of entities* *Weak entity: depends on the existence of another entity*	*One-to-one relationship* *One-to-many relationship* *Many-to-one relationship* *Many-to-many relationship*
Attributes: entities are represented by means of their properties *Types: Simple, composite, derived, single-value, multi-value, stb*	
Relationships: diamond-shaped box, Degree of relationship: Binary = degree 2, ternary = degree 3, n-ary	

ER Modeling Concepts

Specialization	Generalization
	

Data structure diagrams (DSDs)

A simplified representation of how data is stored and connected in a database system.

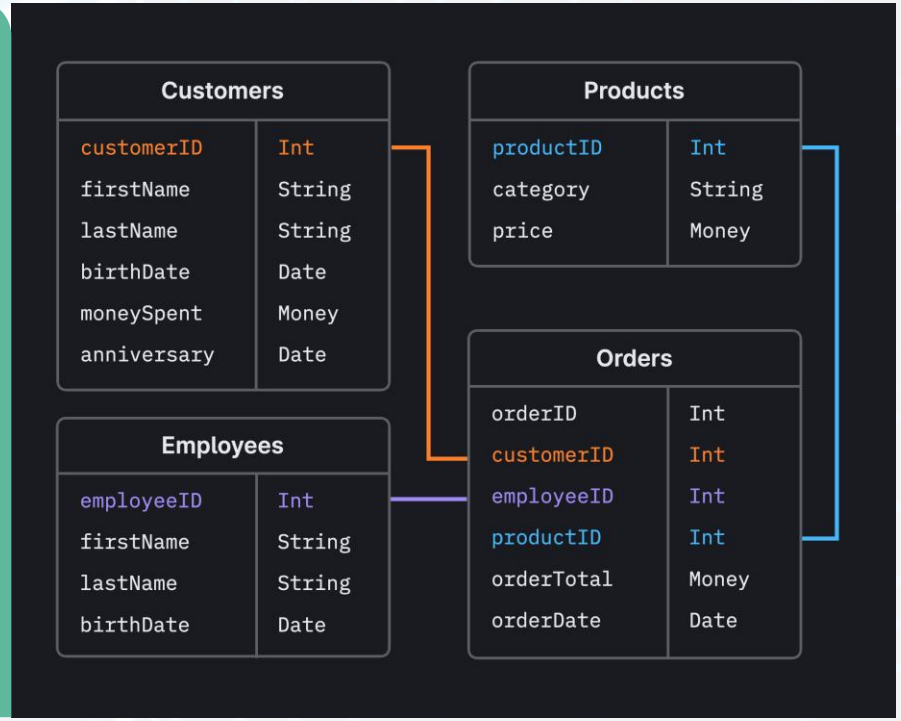
Focus:
- tables (structures)
- keys (connections)
- data organization

ER Model → conceptual view (what exists)
DSD → structural view (how it is stored)

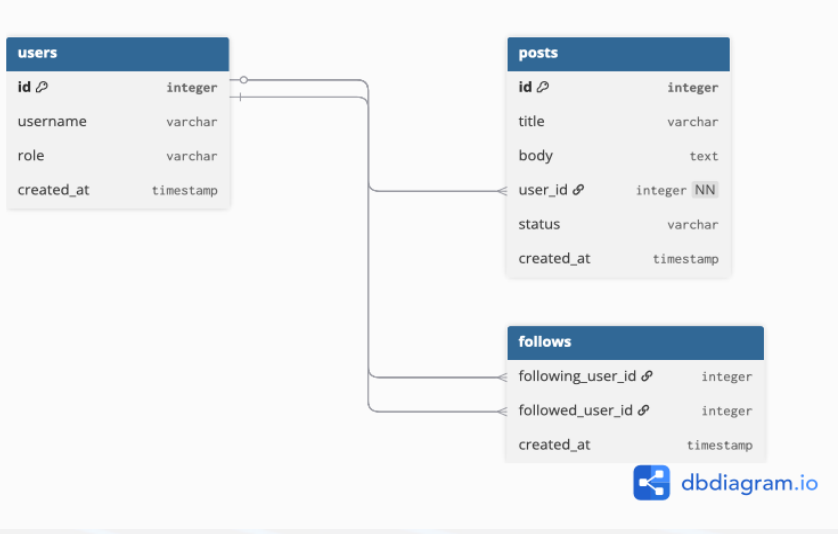
ER: **Person** — has — **Student**

DSD: **Student**.*person_id* → **Person**.*person_id*

https://planetscale.com/blog/schema-design-101-relational-databases



https://dbdiagram.io/



Data types

1. String Data Types

Data type	Description	Max char length	Storage
char(n)	Fixed-length non-Unicode character data (n must be between 1 and 8000)	8,000	n bytes (uses one byte for each character)
varchar(n)	Variable-length non-Unicode character data (n must be between 1 and 8000)	8,000	n bytes + 2 bytes
varchar(max)	Variable-length non-Unicode character data		up to 2 GB
nchar(n)	Fixed-length Unicode character data (n must be between 1 and 4000)	4,000	2 * n bytes (uses two bytes for each character)
nvarchar(n)	Variable-length Unicode character data (n must be between 1 and 4000)	4,000	2 * n bytes + 2 bytes (uses two bytes for each character)
nvarchar(max)	Variable-length Unicode character data		up to 2 GB
binary(n)	Fixed-length binary data (n must be between 1 and 8000)	8,000	n bytes
varbinary(n)	Variable-length binary data (n must be between 1 and 8000)	8,000	actual length of data entered + 2 bytes
varbinary(max)	Variable-length binary data	2GB	

2. Numerical Data Types

Data type	Description	Storage
bit	Integer that can be 0, 1, or NULL	
tinyint	Allows whole numbers from 0 to 255	1 byte
smallint	Allows whole numbers between -32,768 and 32,767	2 bytes
int	Allows whole numbers between -2,147,483,648 and 2,147,483,647	4 bytes
bigint	Allows whole numbers between -9,223,372,036,854,775,808 and 9,223,372,036,854,775,807	8 bytes
decimal(p,s)	Fixed precision and scale numbers.Allows numbers from -10^38 +1 to 10^38 –1. The p parameter indicates the maximum total number of digits that can be stored (both to the left and to the right of the decimal point). p must be a value from 1 to 38. Default is 18. The s parameter indicates the maximum number of digits stored to the right of the decimal point. s must be a value from 0 to p. Default value is 0	5-17 bytes
numeric(p,s)	Fixed precision and scale numbers.Allows numbers from -10^38 +1 to 10^38 –1. The p parameter indicates the maximum total number of digits that can be stored (both to the left and to the right of the decimal point). p must be a value from 1 to 38. Default is 18. The s parameter indicates the maximum number of digits stored to the right of the decimal point. s must be a value from 0 to p. Default value is 0	5-17 bytes
smallmoney	Monetary data from -214,748.3648 to 214,748.3647	4 bytes
money	Monetary data from -922,337,203,685,477.5808 to 922,337,203,685,477.5807	8 bytes
float(n)	Floating precision number data from -1.79E + 308 to 1.79E + 308.The n parameter indicates whether the field should hold 4 or 8 bytes. float(24) holds a 4-byte field and float(53) holds an 8-byte field. Default value of n is 53.	4 or 8 bytes
real	Floating precision number data from -3.40E + 38 to 3.40E + 38	4 bytes

Data types

3. Date and Time Data Types

Data type	Description	Storage
date	Store a date only. From January 1, 0001 to December 31, 9999	3 bytes
time	Store a time only to an accuracy of 100 nanoseconds	3-5 bytes
datetime	From January 1, 1753 to December 31, 9999 with an accuracy of 3.33 milliseconds	8 bytes
datetime2	From January 1, 0001 to December 31, 9999 with an accuracy of 100 nanoseconds	6-8 bytes
smalldatetime	From January 1, 1900 to June 6, 2079 with an accuracy of 1 minute	4 bytes
datetimeoffset	The same as datetime2 with the addition of a time zone offset	8-10 bytes
timestamp	Stores a unique number that gets updated every time a row gets created or modified. The timestamp value is based upon an internal clock and does not correspond to real time. Each table may have only one timestamp variable	

Data types

4. Other Data Types

Data type	Description
sql_variant	Stores up to 8,000 bytes of data of various data types, except text, ntext, and timestamp
uniqueidentifier	Stores a globally unique identifier (GUID)
xml	Stores XML formatted data. Maximum 2GB
cursor	Stores a reference to a cursor used for database operations
table	Stores a result-set for later processing

Structured Query Language

What language do we use to communicate with the database?

- SQL is a standard language for managing and manipulating relational databases.
  Structured - *refers to the organized format of data in tables with rows and columns.*
  Query - *refers to the ability to ask questions and retrieve specific information from the database.*
  Language - *refers to the syntax and rules used to write SQL statements.*

SQL is a declarative language:
- *we specify **what** we want*
- *not **how** to compute it*

The Most Important SQL Commands

Data type	Description	Example
CREATE	Creating objects (database, schema, table, index, references)	CREATE DATABASE <dbname>; CREATE SCHEMA <schema_name>; CREATE TABLE <table_name> (<col1name><col1type><col1attributes> …);
DROP	Dropping objects	DROP TABLE <table_name>;
INSERT INTO	Addig/manipulating data	INSERT INTO <table_name> VALUES (<field1value>… <fieldnvalue>);
UPDATE	Addig/manipulating data	UPDATE <table_name> SET <tabley_name> = <fieldy_value> WHERE <tablex_name> = <fieldx_value>;
DELETE	Remove data	DELETE FROM <table_name>; DELETE FROM <table_name> WHERE <table_name> = <field_value>;
SELECT	Retrieve data	

Today's focus

The Most Important SQL Commands

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COMMAND	Description
CREATE INDEX	Creates a non-unique index (duplicate values are allowed)
CREATE UNIQUE INDEX	Creates a unique index (duplicate values are not allowed)

Cím: Fénykép, Készítette: Ismeretlen a készítő, licenc: CC BY-SA

Today's focus

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Today's focus

BASICS
- SELECT
- FROM
- DISTINCT
- WHERE
- NULL/IS NULL
- AND / OR / NOT
- Arithmetic operators (+, -, *, /)
- Comparison operators (=, <>, <, >, <=, >=)

SORTING
- ORDER BY ASC / DESC
- Multiple column sorting
- Sorting by expressions

FILTERING
- WHERE
- BETWEEN
- IN / NOT IN
- LIKE / ILIKE
- IS NULL / IS NOT NULL
- CASE WHEN
- EXIST / NOT EXIST
- Subqueries in WHERE

AGGREGATING/ GROUPING
- COUNT()
- SUM()
- AVG()
- MIN()
- MAX()
- GROUP BY
- HAVING

WINDOW FUNCTION
- OVER() PARTITION BY
- ORDER BY
- ROW_NUMBER()
- RANK(), DENSE_RANK()
- LAG()
- LEAD()
- SUM(), AVG()

STRING MANIPULATION
- CONCAT()
- ||
- UPPER() , LOWER()
- TRIM(), LTRIM(), RTRIM()
- SUBSTRING(), LEFT(), RIGHT()
- REPLACE()
- LEN()
- COALASCE()
- NULLIF()

CTEs / SUBQUERYS
- WITH CTE AS ()
- Subquery in SELECT / FROM / WHERE
- Recursive CTE
- Correlated subquery

SET OPERATIONS
- UNION
- UNION ALL
- INTERSECT
- EXCEPT / MINUS

DATA CLEANING
- DISTINCT
- COALESCE()
- CAST()
- TRIM()
- REPLACE()
- LOWER(), UPPER()
- CASE WHEN
- NULLIF()

DATE MANIPULATION
- DATE / TIMESTAMP
- NOW(), CURRENT_DATE
- DATEADD()
- EXTRACT()
- CAST()
- DATE_TRUNC()
- YEAR(), MONTH(), DAY()

JOINS
- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL OUTER JOIN
- CROSS JOIN
- SELF JOIN

SQL Query Logical Order

SELECT *column_a, column_b*

FROM *t1*

JOIN *t2*

ON *t1.column_a = t2.column_b*

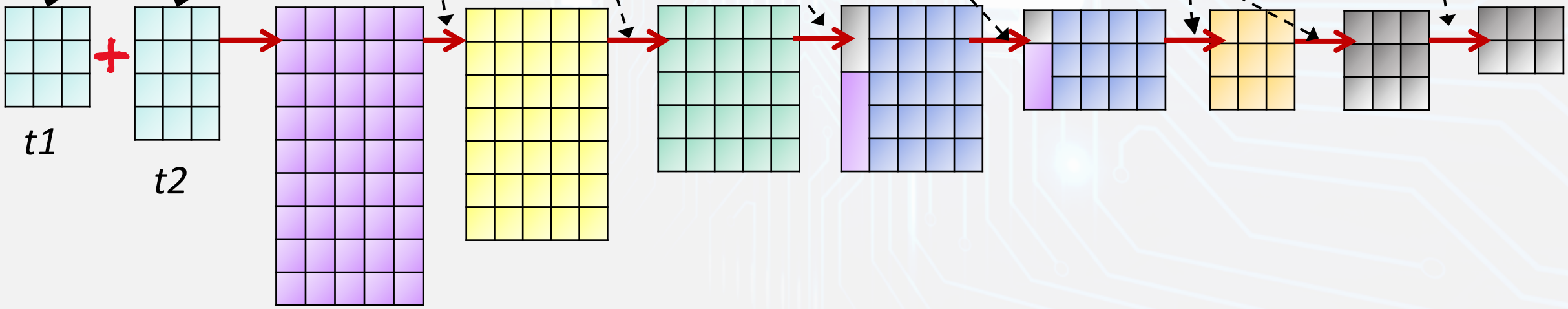
WHERE *constraint_expression*

GROUP BY *column*

HAVING *constraint_expression*

ORDER BY *column ASC / DESC*

LIMIT *count;*



JOIN Types

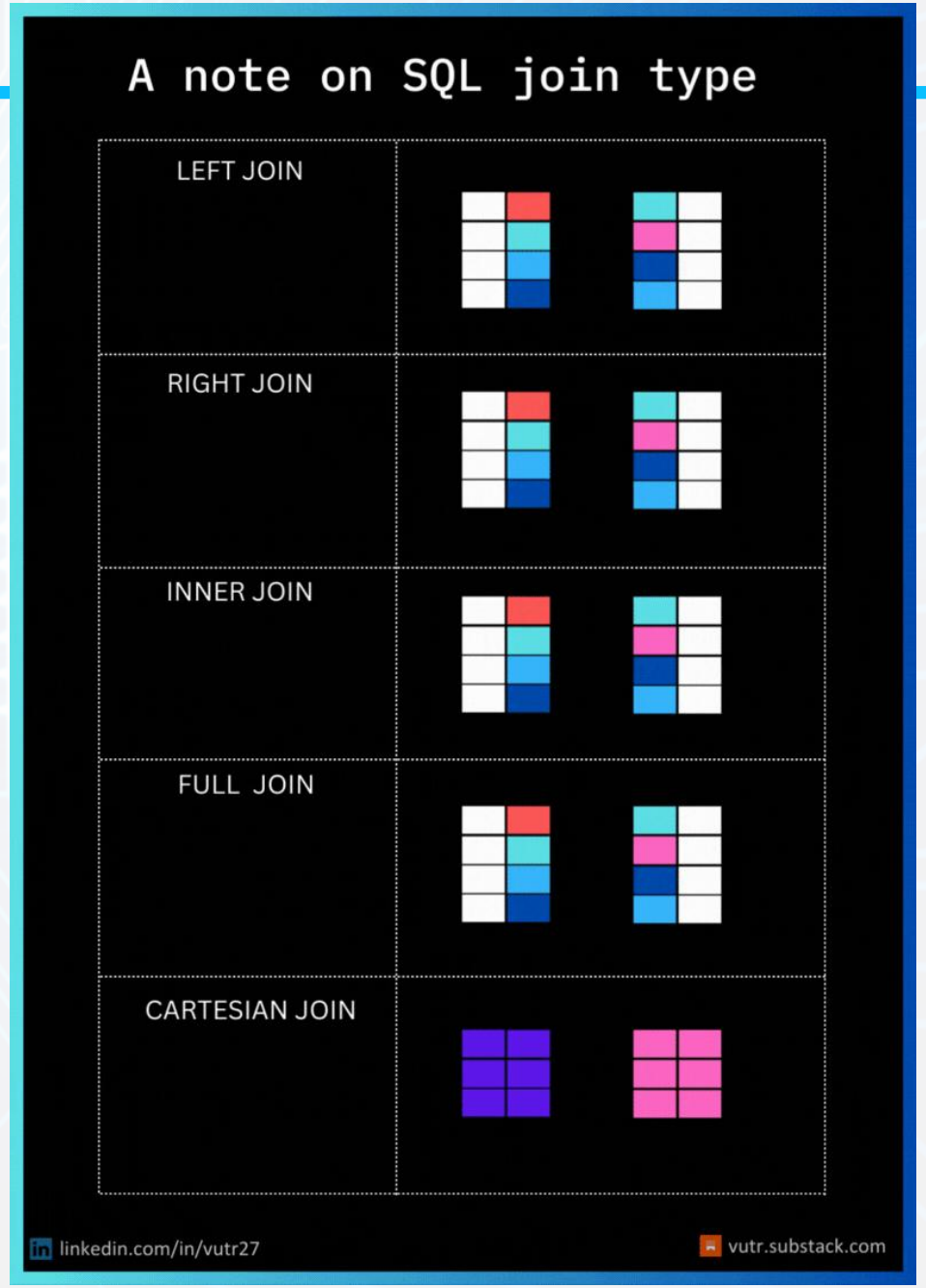
https://datalemur.com/blog/sql-joins-infographics

JOIN_TYPE (X): LEFT, RIGHT, INNER, FULL

```
SELECT columns
FROM table1
X JOIN table2
ON table1.common_column = table2.common_column;
```

CARTESIAN (CROSS) JOIN

```
SELECT columns
FROM table1
CROSS JOIN  table2;
```



Metadata and data management

Metadata describes the data:

- *column meanings*
- *units*
- *data types*
- *origin of data*

Example:

COLUMN	MEANING	UNIT
Temperature	Air temperature	°C
Date	Measurement date	YYYY-MM-DD

In scientific databases, metadata is essential for:

- *interpretation*
- *reproducibility*
- *data sharing*

Data without metadata is often meaningless.

Keep in Mind That...

1. **SQL keywords are NOT case sensitive:** *SELECT = select = SeLeCt*

2. **6. Whitespace and formatting do NOT affect execution:** → *but improve readability*

3. **BUT! Table and column name sensitivity depends on the database system and configuration (collation):** *depending on the system, PostgreSQL vs MySQL vs SQLite → big differences*

4. **SQL is declarative, not procedural:** → *you say WHAT you want, not HOW to get it*

5. **Queries do NOT modify data unless explicitly stated:** *SELECT is safe*

6. **Statements are typically terminated with a semicolon (required in some environments)**

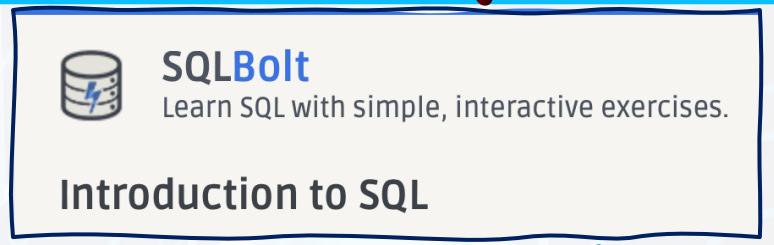
7. **NULL is NOT a value** → *comparisons with NULL require IS NULL / IS NOT NULL*

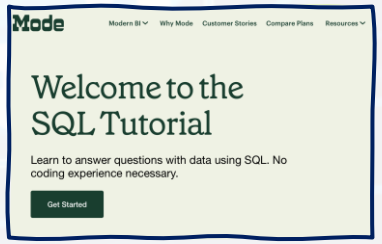
8. **SQL is written top-down, but executed in a different logical order**

Top SQL learning resources

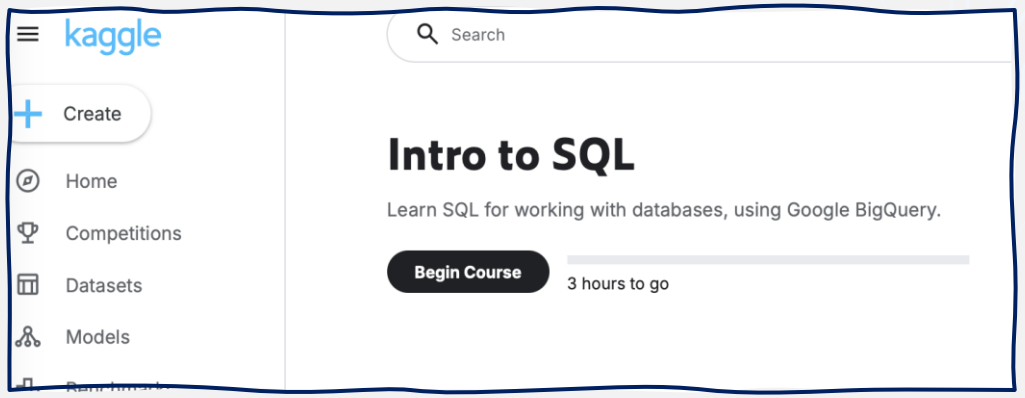
... in my opinion

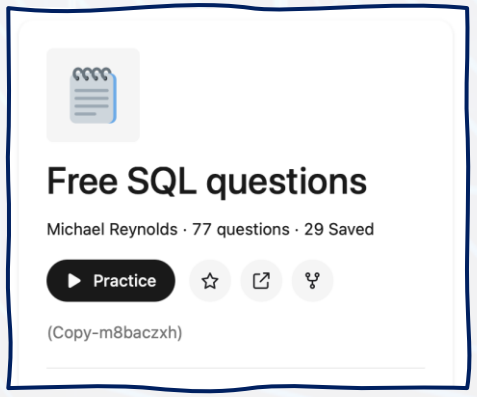
- **Mode -** *https://mode.com/sql-tutorial/sql-in-mode*
- **SQLBolt -** *https://sqlbolt.com*
- **Khan Academy -** *https://www.khanacademy.org/computing/computer-programming/sql*
- **W3Schools -** *https://www.w3schools.com/sql/*
- **Kaggle -** *https://www.kaggle.com/learn/intro-to-sql*
- **Leetcode:** *https://leetcode.com*
- **Youtube: e.g.** *https://youtu.be/h0nxCDiD-zg?si=alyTJoy1d6PBAAB5*, *https://youtu.be/OT1RErkfLNQ?si=W5txFtbbBdsr_AMB*

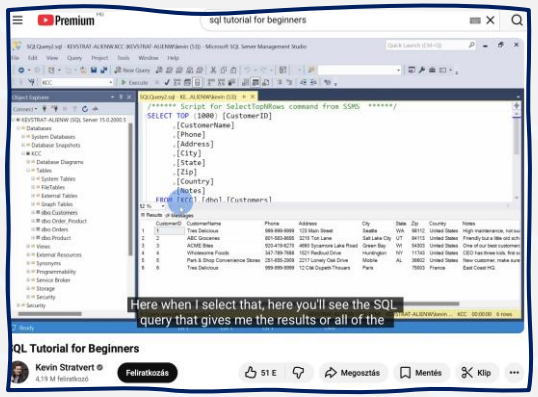




Khan Academy



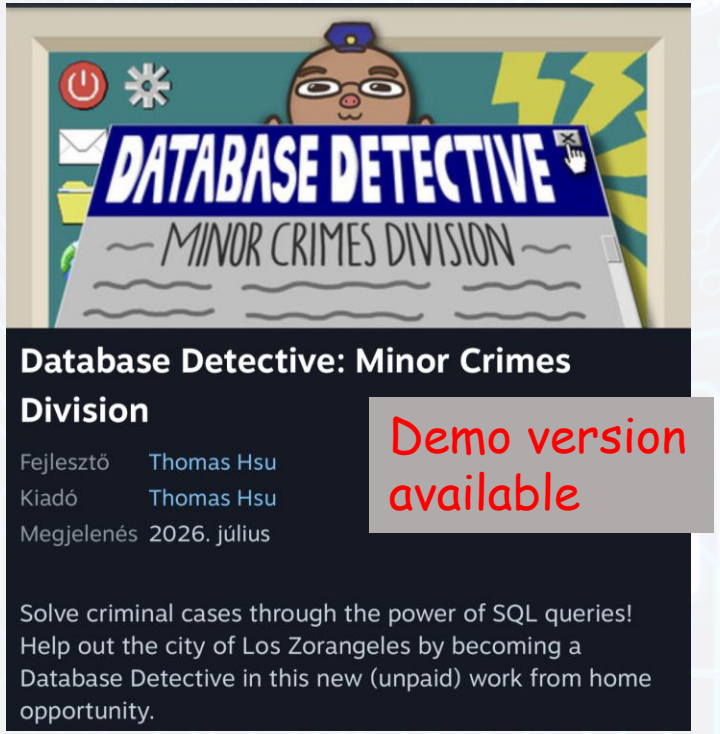


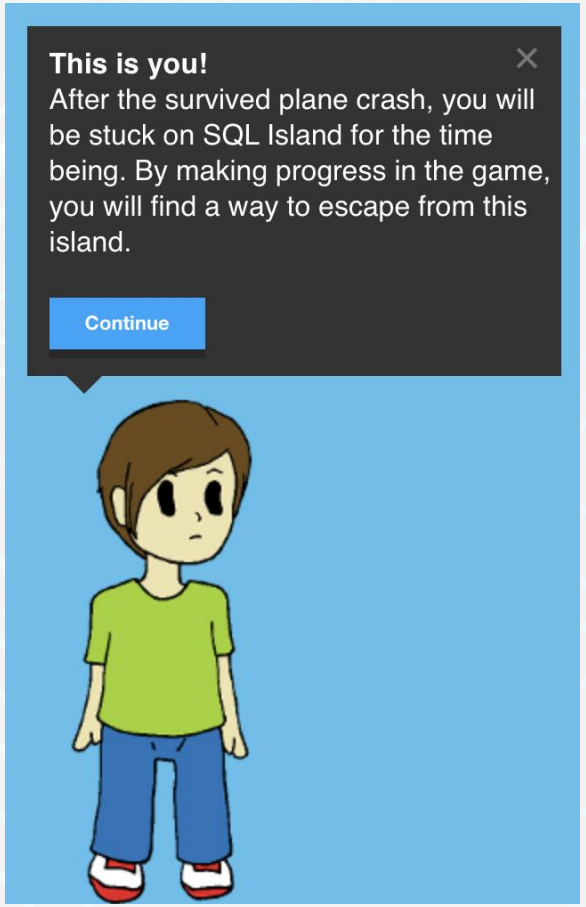


Games to learn SQL

https://mystery.knightlab.com

https://store.steampowered.com/app/3950130/Database_Detective_Minor_Crimes_Division/





https://sql-island.informatik.uni-kl.de

5 SQL Games to Master Database Skills in 2025

https://www.sqlnoir.com/blog/games-to-learn-sql